GNR-602

Advanced Methods in Satellite Image Processing

Comparative Analysis of Quadtree and K-Means Segmentation on Wavelet-Denoised Satellite Imagery

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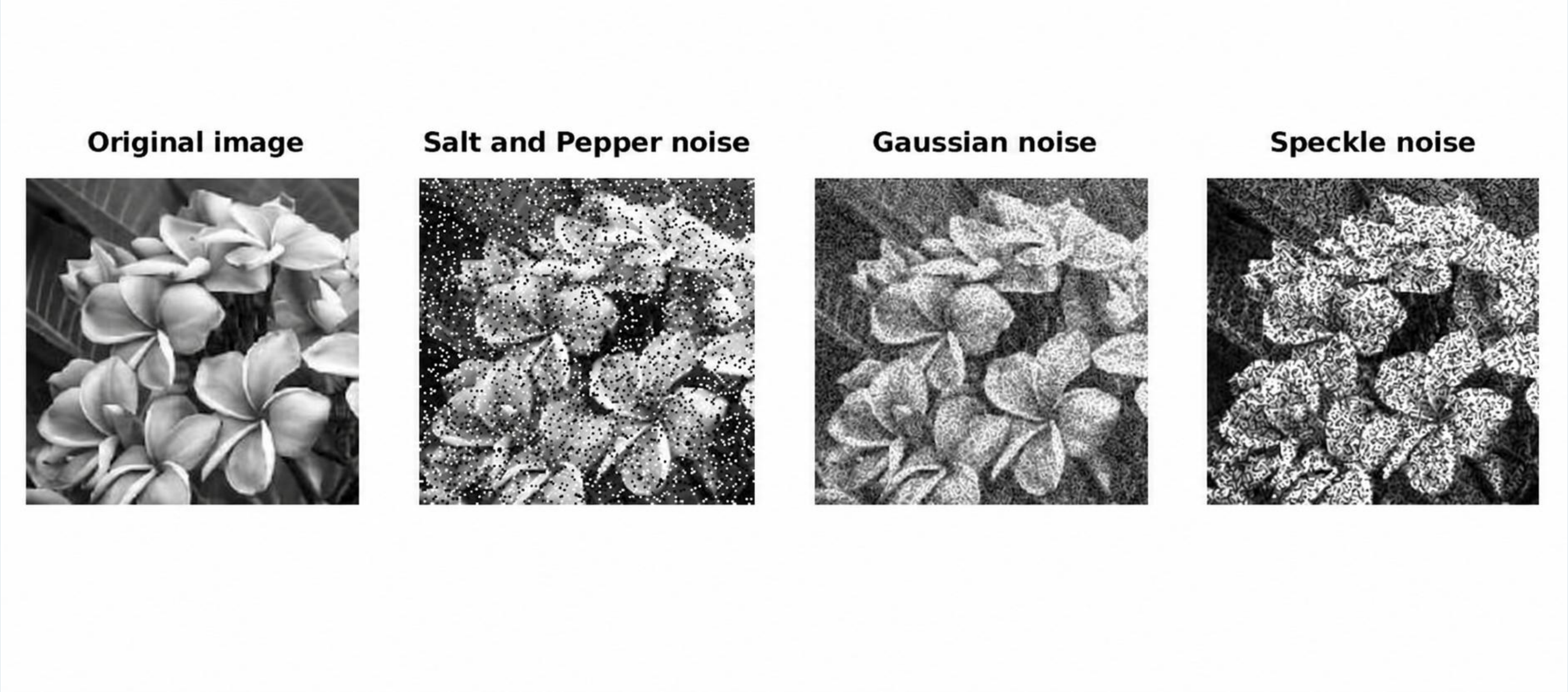
Problem Statement

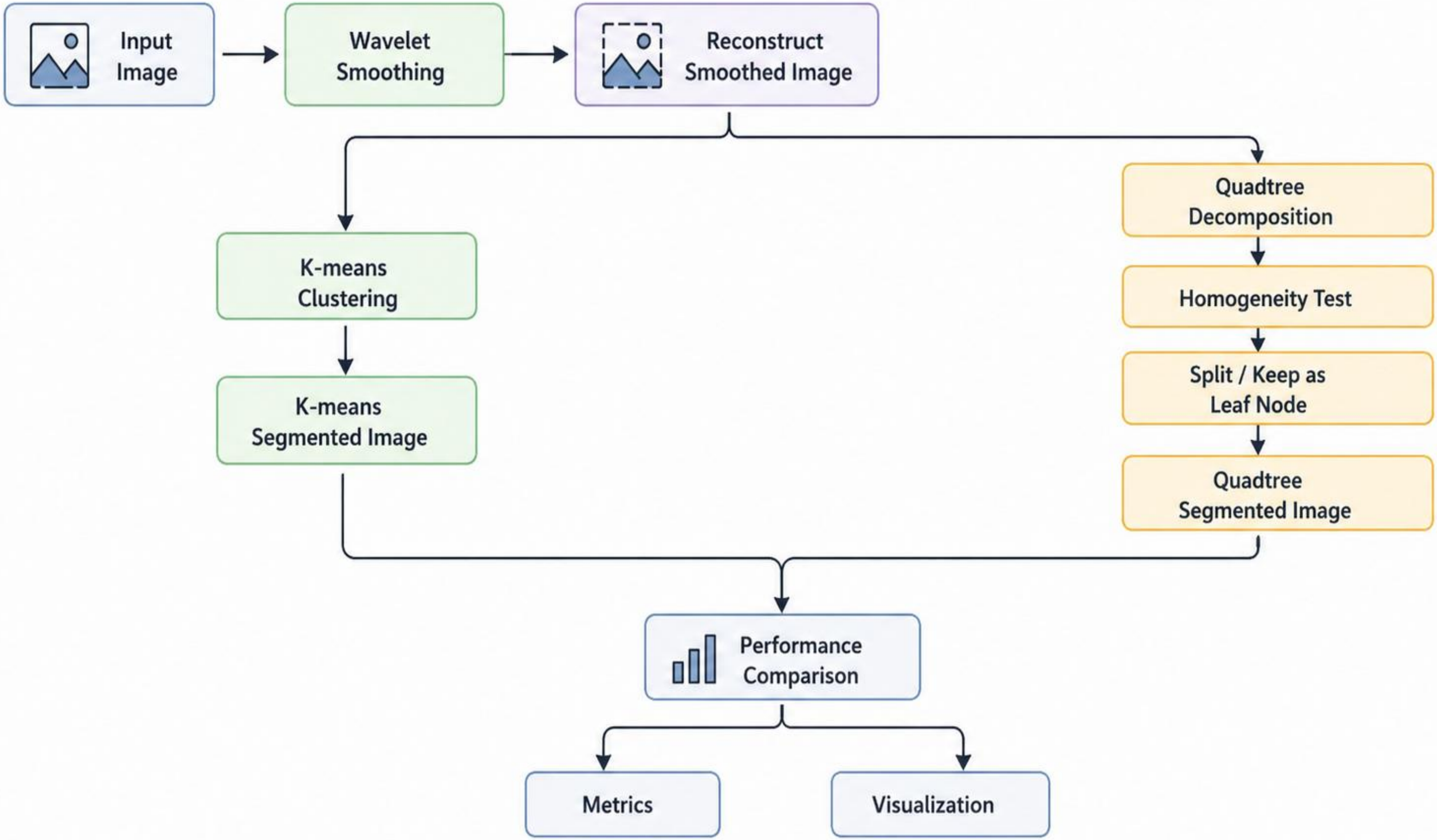
“ Noise degrades segmentation accuracy, and conventional methods fail to capture spatial structure. This project addresses the need for combining smoothing with segmentation for better region extraction.”



Objectives

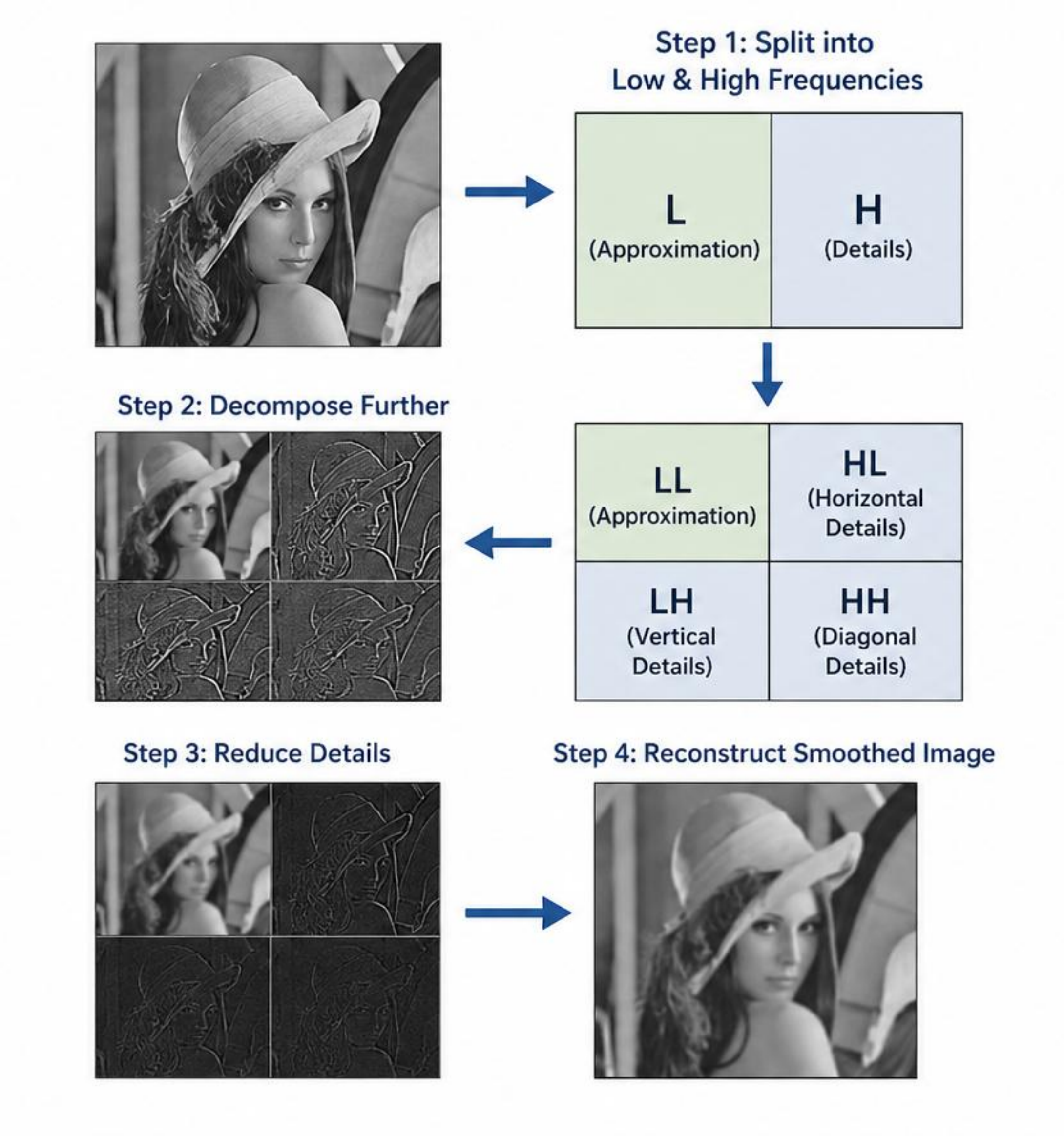
- Use Daubechies wavelets (**db4 and db6**) for image smoothing.
- Reduce noise while preserving image details and structures.
- Perform **quadtree-based segmentation** using homogeneity criteria.
- Compare segmentation performance with **K-means clustering**.
- Evaluate results using **standard segmentation metrics.**

PIPELINE ARCHITECTURE



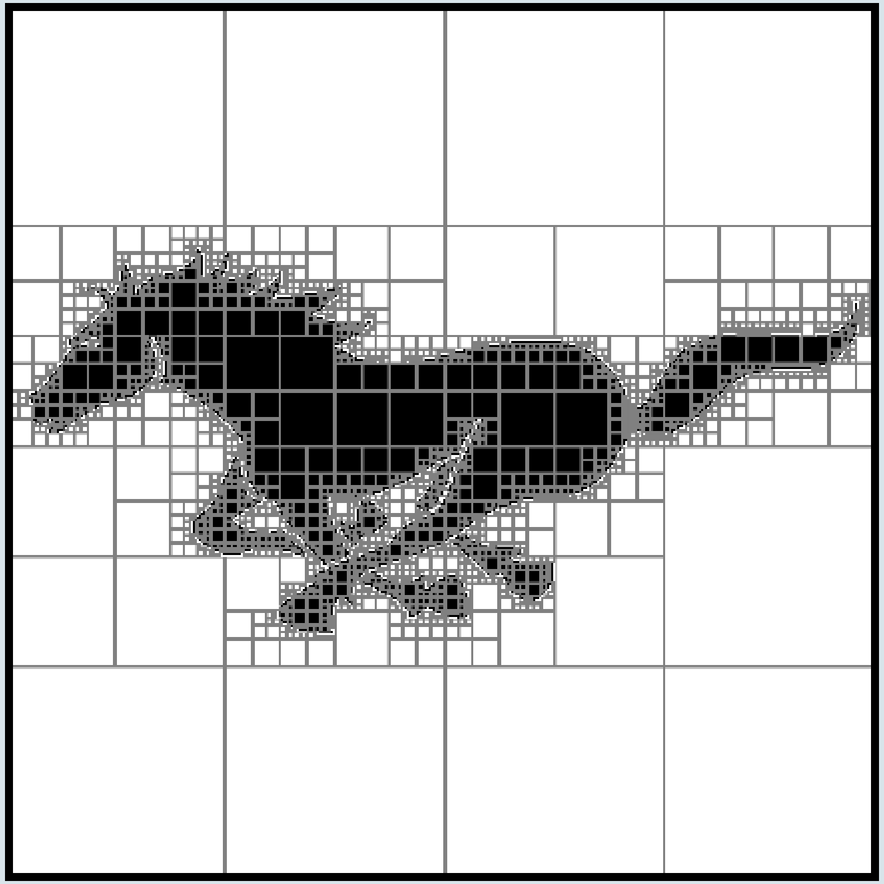
Wavelet Smoothing

- Image is split using **Daubechies wavelets (db4 / db6)**
- Decomposition separates image into two parts:
  - **Approximation (low frequency**
  - **Details (high frequency**
- Approximation part is **kept unchanged**
- Detail coefficients are **scaled down (reduced)**
- db4 = moderate smoothing, db6 = more smoother result
- Reducing details removes noise but preserves structure
- Final image becomes smoother and cleaner for segmentation



Quadtree Segmentation

- Image is divided into **square regions based on pixel similarity**
- Start with the **entire image and check homogeneity**
- Non-uniform regions are **recursively split into 4 parts**
- Process continues until regions become **uniform or reach minimum size**
- Final blocks represent **segmented regions of the image**
- Large blocks indicate **smooth areas**, small blocks indicate **detailed regions**



Quadtree Segmented Image Construction:

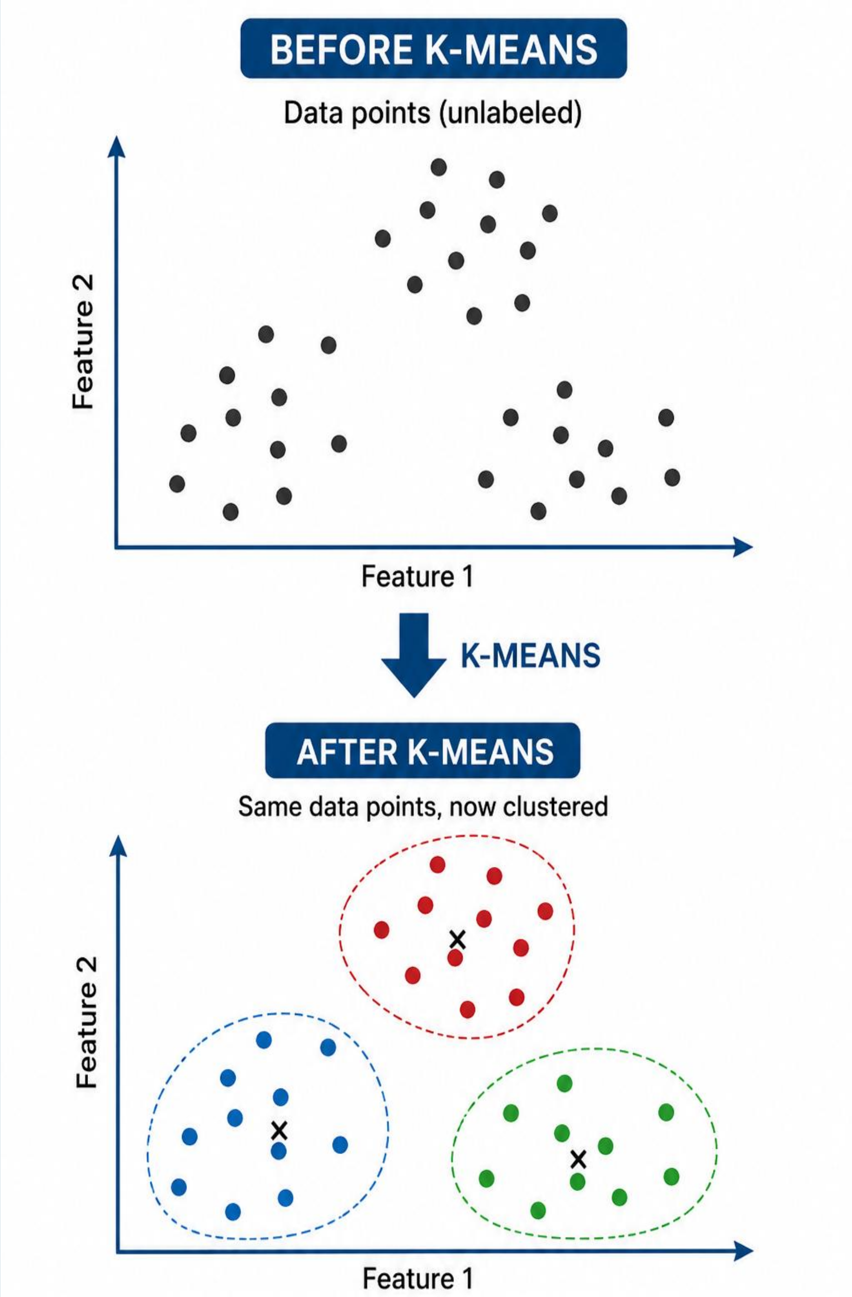
- Use the **final leaf blocks (homogeneous regions)**
- Compute **mean intensity for each block**
- Assign this value to **all pixels within the block**
- This directly forms the **segmented image**

key parameters:

Parameter	Description	Formula
Homogeneity Threshold (T)	Decides whether a block should be split	$\sigma > T$
Standard Deviation (σ)	Measures variation inside a block (homogeneity check)	$\sigma = \sqrt{\frac{1}{N}\sum(x_i - \mu)^2}$
Mean Intensity (μ)	Represents each region (final pixel value)	$\mu = \frac{1}{N}\sum x_i$

K-Means Segmentation

- Groups pixels based on **intensity similarity**
- Number of clusters (**K**) is predefined
- Each pixel assigned to **nearest cluster center**
- Cluster centers updated **iteratively**
- Produces segmented image with **K regions**



Parameters(k-means):

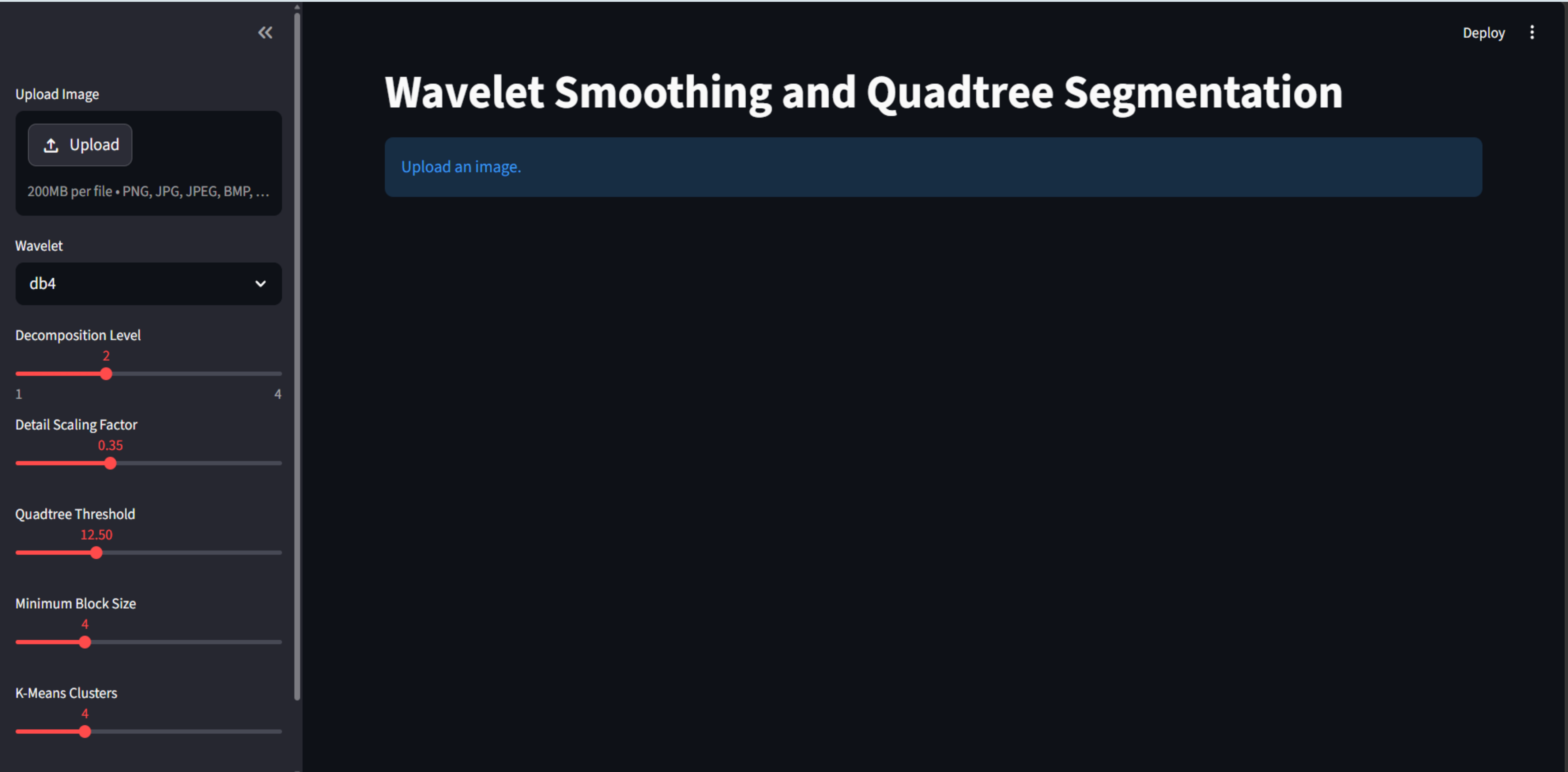
Parameter	Description	Formula
Number of Clusters (K)	Number of regions to segment the image into	--
Cluster Centers (μ_k)	Representative intensity of each cluster	$\mu k = (1 / Nk) * \Sigma xi$
Distance Metric	Assigns pixel to nearest cluster center	$d = \sqrt{[(x - \mu k)^2]}$
Objective Function (Loss)	Minimizes intra-cluster variance	$J = \Sigma (x - \mu k)^2$
Initialization	random	--

Metrics

Metric	Describes	Formula
MSE (Mean Squared Error)	Difference between original and segmented image	$MSE = (1/N) * \Sigma (I - \hat{I})^2$
Intra-region Variance	Variation within each segment (homogeneity)	$\sigma^2 = (1/N) * \Sigma (x - \mu)^2$
Number of Regions	Total segments formed	--
Difference Map	Visual difference between methods	--

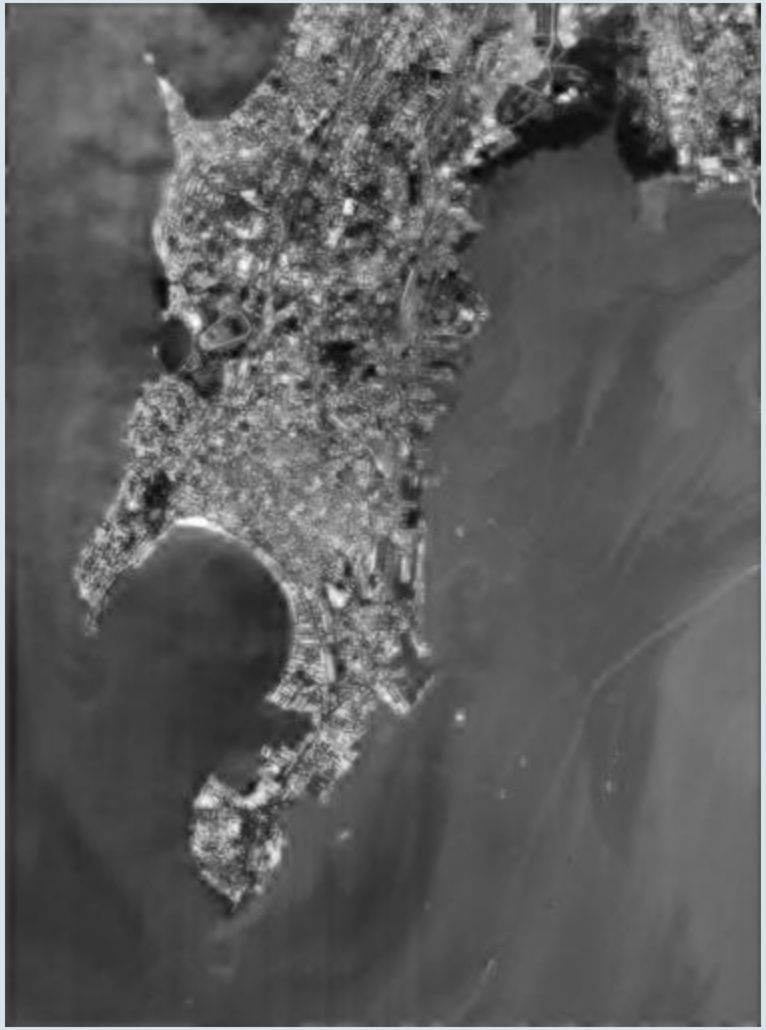
"Lower MSE → better reconstruction"
"Lower variance → more uniform regions"

UI - Streamlit

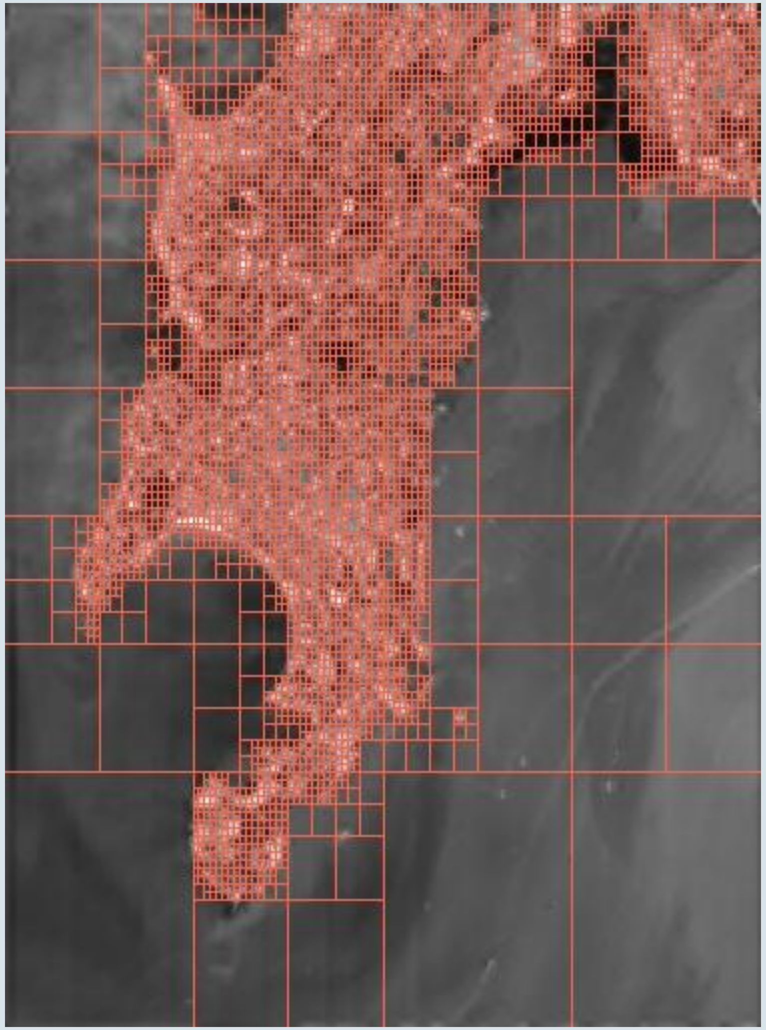


Results

Original Image

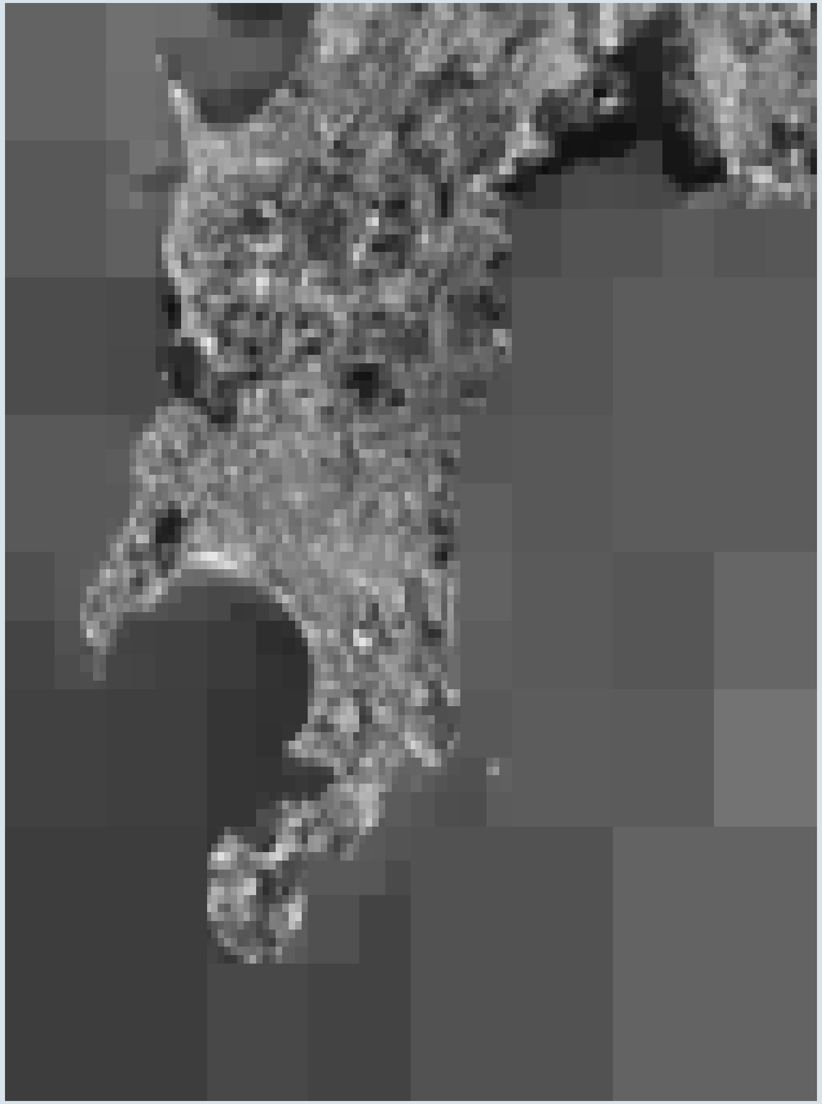


Wavelet Smoothed Image

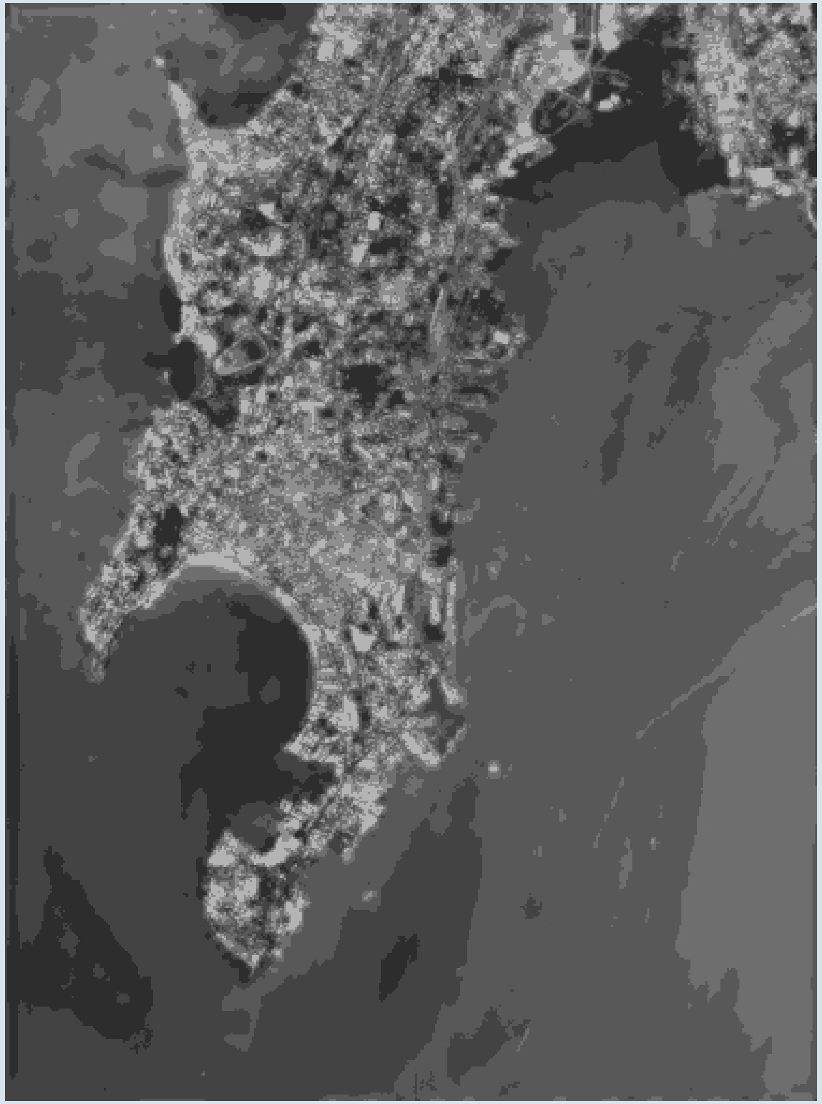


Quadtree segmentation

Comparison b/w Quadtree & K-means Segmented Images:



Quadtree Segmented Image



K-means Segmented Image

Metrics: Quadtree vs K-means

Metrics

Metric	Quadtree	K-Means
MSE	171.50	56.83
Intra-region Variance	171.43	56.48
Number of Segments	5443	6

Observations:

Wavelet smoothing reduces noise while preserving key structures

Quadtree adapts to image content → large blocks (smooth), small blocks (detailed)

Quadtree segmentation captures spatial structure effectively but often leads to over-segmentation, especially in high-detail areas

K-means produces **coarse, uniform regions** with fixed clusters

K-means **does not consider spatial information**, so distant but similar pixels may be grouped together

Quadtree boundaries follow image structure, K-means boundaries are less aligned

Difference map highlights variations in complex regions

Thank you